

Question:
Suppose that there are two firms in the market facing no costs of production and a demand curve given by $Q = 150 - P$ for their identical products. Suppose the two firms choose prices simultaneously as in the Bertrand model. Compute the prices in the nash equilibrium.

Answer Choices:

- A. 30
- B. 50
- C. 0
- D. 25
- E. 150
- F. 75
- G. 90
- H. 120
- I. 100
- J. 60

Reasoning + Answer:

</tool_call> reasoning process here :

In the Bertrand model with two firms producing identical products and zero costs , the Nash equilibrium occurs when both firms set their prices equal to the marginal cost of production . Since the problem states that there are no costs of production , marginal cost is zero .

In a Bertrand model with identical products and zero marginal cost , each firm has an incentive to undercut the other 's price slightly to capture the entire market . This continues until the price reaches zero , as any positive price could be undercut by the rival to capture all consumers .

Therefore , in the Nash equilibrium , both firms will set price equal to zero .

Thus , the Nash equilibrium price is zero .

< answer > 0 </answer >

